MANIPAL ACADEMY OF HIGHER EDUCATION

IV SEMESTER BTECH (CSE)
MID TERM EXAMINATION - MARCH 2025

EMBEDDED SYSTEMS [CSE 2223]

Marks: 30 **Duration: 90 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) Identify the amount of memory occupied within the address range 0x60000000 – 0x6001FFFF for SRAM. (0.5)

0x1FFFF KB 128 MB 128 B 128 KB

2) The ability to shift or rotate in the same instruction along with other operations is performed with the help of _________ (0.5)

Switching circuit | Barrel shifter | Integrated Switching circuit | Multiplexer circuit

3) Identify the operation that does not exist for the ADD instruction in RISC. (0.5)

Register to Register | Memory to Memory | Immediate to Register | None of the given options

4) The BEQ instruction is used ______ (0.5)

To check the inequality condition between operands and then branch | To check if the operand is greater than the condition value and then branch | To check if the flag Z is set to 1 and then causes branch | None of the other options

5) The register which holds the address of the instruction to be executed next is ________ (0.5)

R15 R14 R13 R12

6)	Identify which of the following address/es is/are word aligned (i) 0x52000068 (ii) 0x6600082 (iii) 0x18000014 (iv) 0x1200004A i) and iii) only ii) and iv) only ii) and iii) only i), ii), iii) and iv)	(0.5)
7)	Identify the ARM instruction that will conditionally move data from R1 to R2 only if the Zero (Z) flag is set. MOV R2, R1 MOVNE R2, R1 MOVEQ R2, R1 CMN R1, R1	(0.5)
8)	What is the result of the following ARM instruction? EOR R1, R2, #0xFF R1 = R2 AND 0xFF R1 = R2 OR 0xFF R1 = R2 NOR 0xFF R1 = ~R2	(0.5)
9)	Identify the statement(s) that can be used to clear the values in port pins 1.11 to 1.16? (I) LPC_GPIO0->FIOSET=0x1F800 (II) LPC_GPIO0->FIOCLR=0x1F800 (III) LPC_GPIO1->FIOPIN=0x0 (IV) LPC_GPIO1->FIOCLR=0x1F800 I and II only II and III only III and IV only I and IV only	(0.5)
10)	The purpose of the FIOxMASK register is _________. To store the pin's logic state To enable or disable access to specific GPIO pins To configure alternate functions To generate an interrupt	(0.5)

DESCRIPTIVE

Answer all the questions.

Assume any missing information suitably.

11)	Develop an ARM Assembly Language program that performs the following operations: (i) Read a 16-bit value from memory location 0x20001000. (ii) Swap the upper and lower bytes of this value. (iii) Store the modified value back at 0x20001004. Provide comments to explain your logic.	(4)
12)		(4)

Develop an Embedded C Program to implement 4-bit binary up counter in the most significant 4 LEDs of the LED block in LPC1768 when the switch SW2 is pressed. Assume that the LEDs are connected to the port pins from P0.15 to P0.22 and the switch SW2 is connected to P2.0.

13) Develop an Embedded C Program to display the text "CSE" in the seven-segment display (SSD) unit continuously. The text should be displayed in units U11, U10 and U9 of the SSD unit. Assume that the eight segments of all the SSD units are connected to the port pins P0.4 to P0.11 and the SSD units can be enabled one by one using the port pins P0.15 to P0.18. (4)

14) Identify the memory addresses referenced by each of the instructions in the following code snippet. The instructions are executed sequentially and the initial values in the registers are as follows: R2 = 0X2000, R3 = 0XFF, and R4 = 0X1000. (3)

LDR R1, [R2, R3, LSL #2]

STR R1, [R4, R3]

LDRH R1, [R2, #4]

STRH R1, [R4, #2]!

LDRB R1, [R4], #1

STRB R1, [R2].

15) Discuss the functionalities of FIOMASK register with an example. (3)

16) Assume the values in Registers R5 – R8 are 0x205 to 0x208 respectively. The value of the Stack Pointer (R13) is 0x40002000. Use appropriate instructions to perform push and pop operations on fully ascending and empty descending stacks. Explain the operation with neat diagrams. (3)

I. Store the values from R5 – R8 onto the stack.

II. Restore the values into R5 – R8 after the push operation.

Note: (Use LDM and STM with appropriate suffixes other than EA, FA, ED, and FD suffixes).

17) Explain the syntax and working of the instruction RRX in ARM assembly language. Show the steps and the result while performing RRX on the data 0x85. (2)

18) Compare the AND and BIC instructions with examples. (2)